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| **Fiscal Year:** | FY 2020 | **Task Last Updated:** | FY 12/23/2019 |
| **PI Name:** | Voorhees, Peter W. Ph.D. | | |
| **Project Title:** | The Columnar to Equiaxed Transition and Dendrite Fragmentation | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 60208-0893 | **Congressional District:** | 9 |
| **Comments:** | | | |
| **Project Type:** | Flight | **Solicitation / Funding Source:** | 2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station |
| **Start Date:** | 01/01/2017 | **End Date:** | 12/31/2021 |
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| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Gulsoy, Emine Ph.D. ( Northwestern University ) | | |
| **Grant/Contract No.:** | NNX17AD48G | | |
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| **Performance Goal Text:** | | | |

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| **Task Description:** | The shape and size of dendritic grain structures in engineering alloys is central to the resulting properties of the material. For example, equiaxed grain structures tend to provide more isotropic materials properties then materials with columnar grain morphologies. By contrast, in other applications, such as NI-base superalloys, columnar grain structures provide superior high temperature or creep properties. It is thus of critical importance to understand and predict the factors affecting the formation of grain morphology during solidification. The processes controlling the formation of equiaxed and columnar grains during solidification will be investigated using two approaches. The first is to measure the grain structures of Al-Si alloys processed by the Columnar-to-Equiaxed Transition in SOLidification Processing (CETSOL) team that is sponsored by the European Space Agency (ESA), and the second is to examine the effects of solidification on the dendrite fragmentation process. By processing these samples on the International Space Station (ISS), both sedimentation of the dendritic fragments and buoyancy driven convection can be avoided. This will permit an understanding of the physics underlying these processes to be gained, and enable a careful comparison with simulation. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | In nearly all casting processes, the resulting solid consists of an array of grains of different crystallographic orientation. Depending on the processing parameters, columnar or equiaxed grain growth is present, while in others there is a transition from columnar to equiaxed grains (CET). The importance of the CET lies in the clear connection between the properties of the solidified ingots and the grain structure. For example, ingots with elongated grains can have much different creep behavior than those with equiaxed grains. The grains in this case can be a single dendrite, as in the equiaxed region, or a collection of dendrites with the same crystallographic orientation. In many cases, such as in hypoeutectic alloys during cooling a mush of dendrites and liquid is formed. The last to solidify liquid forms a eutectic both between the secondary dendrite arms, but more importantly in equiaxed structures between the grains. This can lead to casting defects, such as hot tearing and intergranular cracking. Thus, the conditions under which an ingot solidifies with a columnar or equiaxed structure can have a major effect on the properties of the resulting part. Moreover, if the conditions under which the CET occurs are well understood, a properly placed CET can be used to tailor the properties of the ingot, as well as the spatial distribution of the intergranular eutectic. |
| **Task Progress:** | The 40 vol% spaceflight samples from the Coarsening in Solid-Liquid Mixtures (CSLM)-4 experiments have been being serial sectioned, segmented, and analyzed. Currently, the 20 vol% spaceflight samples are in the process of being segmented. The segmentation process can be a bottleneck as artifacts in the form of smudges and speckles make it very difficult to accurately segment the dataset computationally, while the size of the dataset makes segmenting by hand infeasible. The exact artifacts vary from sample to sample making some samples much easier to segment than others. For this reason, we turned to machine learning to see if this emerging technology could help fix the issues encountered. Using a convolutional neural network trained on data from CSLM-4, we were able to segment images that were very difficult to analyze in the past. We will be using this approach to analyze the data from the upcoming spaceflight experiment.<br>Ground-based tests have been conducted in a tube furnace to determine the feasibility of the upcoming spaceflight experiment as well as to provide a comparison to the spaceflight experiment. These tests involved coarsening samples with an axial rotation to constantly change the gravitational vector direction experienced by the sample to approximate diffusion-controlled coarsening. During this process, samples underwent a set heating or cooling rate and their microstructures were compared to isothermal samples kept above the eutectic temperature for the same length of time. These rates were partially determined from our previous work with eroding dendritic microstructure reconstructions, as these gave some insight into a maximum temperature that we could expect to reach before the dendritic microstructure melted into many different solid fragments. These experiments will form the basis of comparison to our spaceflight experiment. |
| **Bibliography Type:** | Description: (Last Updated: 01/17/2024) |
| **Articles in Peer-reviewed Journals** | Stan T, Thompson ZT, Voorhees PW. "Optimizing convolutional neural networks to perform semantic segmentation on large materials imaging datasets: X-ray tomography and serial sectioning." Materials Characterization. 2020 Feb;160:110119. https://doi.org/10.1016/j.matchar.2020.110119 , Feb-2020 |